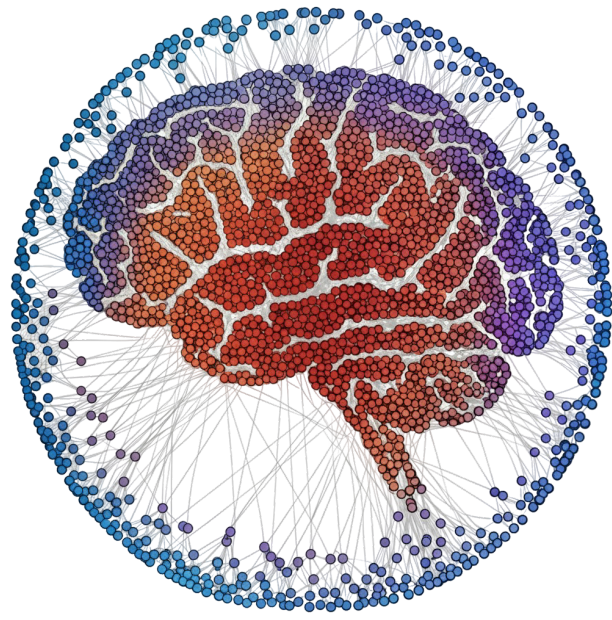




Morse-Seeded Coresets for Structure-Preserving Landmarking on kNN Graphs

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Core claim. A landmark set should preserve both **metric coverage** and **graph-induced structure**. We seed farthest-first traversal using SCCs of a directed density flow on the kNN graph, then test whether this structural prior improves representation of recurrent/boundary regions without sacrificing coverage.

1. Problem, Definitions, and Target

Landmarks / coreset. A landmark set is a small representative subset

$$L \subset V, \quad |L| = m \ll n,$$

used to approximate coverage and graph distances on the full point cloud.

Coverage. We measure the worst nearest-landmark distance:

$$r(L) = \max_{i \in V} \min_{l \in L} \|x_i - x_l\|_2.$$

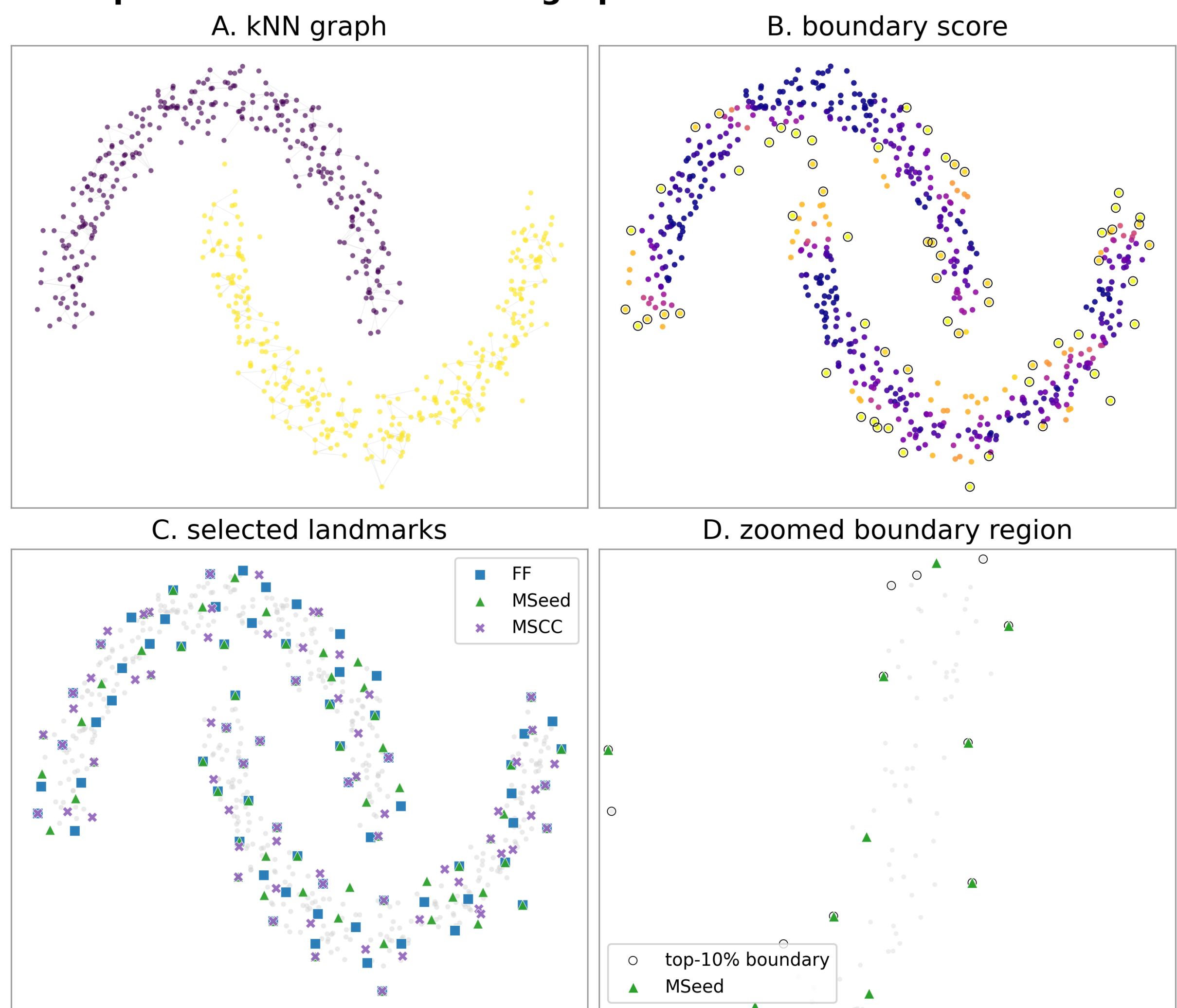
Structure-preserving. Let $d_G(i, j)$ be shortest-path distance on the full weighted kNN graph, and let $\hat{d}_L(i, j)$ be the landmark approximation. We evaluate

$$E_{\text{geo}}(L) = \text{median}_{(i,j)} \left| \frac{\hat{d}_L(i,j) - d_G(i,j)}{d_G(i,j)} \right|.$$

The structural test is not just "low error"; it also asks whether selected points hit high-boundary recurrent regions.

2. Conceptual Method

Conceptual visualization: from graph structure to seeded landmarks



A simple kNN graph before sampling. Black rings show the top 10% SCC-boundary nodes. Morse-guided selection seeds recurrent or boundary regions before global farthest-first fill.

3. Algorithm and Metrics

Density potential.

$$\Phi(i) = \text{distance from } x_i \text{ to its } k\text{-th nearest neighbor.}$$

Directed flow. Add $i \rightarrow j$ when $j \in \text{kNN}(i)$ and

$$\Phi(j) \leq \Phi(i) \quad \text{or} \quad |\Phi(j) - \Phi(i)| \leq \varepsilon.$$

SCCs of G^d act as a coarse combinatorial Morse proxy.

MorseSeeded-FF. Choose boundary-score representatives from large SCCs as seeds S , then run global farthest-first initialized by S to fill the budget m .

Boundary recall at top 10%. Let B_{10} be the top 10% of all nodes by SCC-boundary score. We report

$$R_{10}(L) = \frac{|L \cap B_{10}|}{|B_{10}|},$$

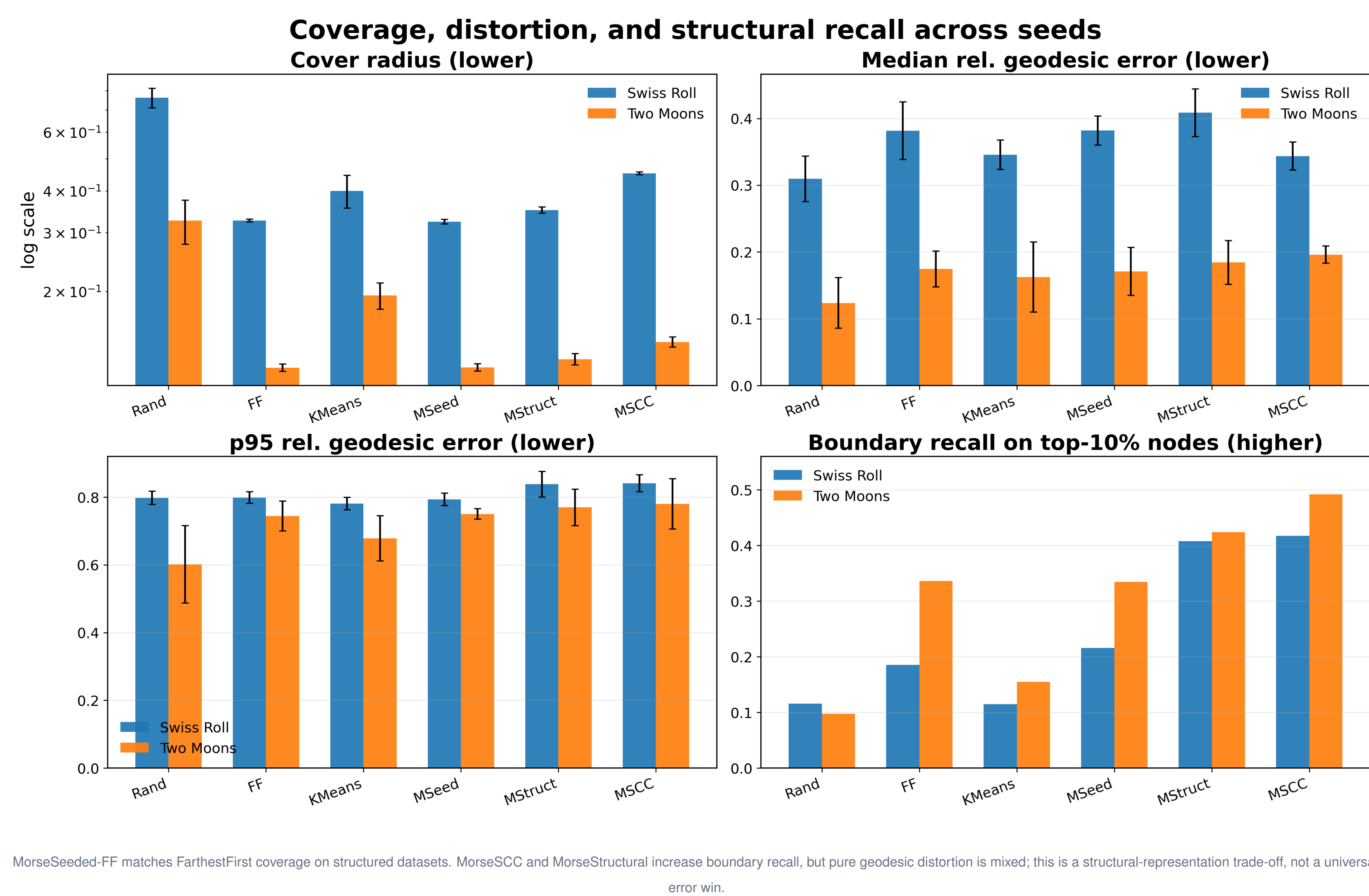
so 10% refers to *nodes ranked by boundary score*, not the coreset size.

4. Main Empirical Summary

Dataset	FF cover	MSeed cover	Best median E_{geo}	Best p95 E_{geo}	Best R_{10}	Interpretation
Swiss Roll	0.326	0.324	Rand 0.310	KMeans 0.781	MSCC 0.417	coverage preserved; structure recalled
Two Moons	0.118	0.118	Rand 0.124	Rand 0.601	MSCC 0.492	strongest boundary signal
Digits PCA	5.257	5.257	KMeans 0.153	KMeans 0.625	MSCC 0.167	real-data signal is weaker

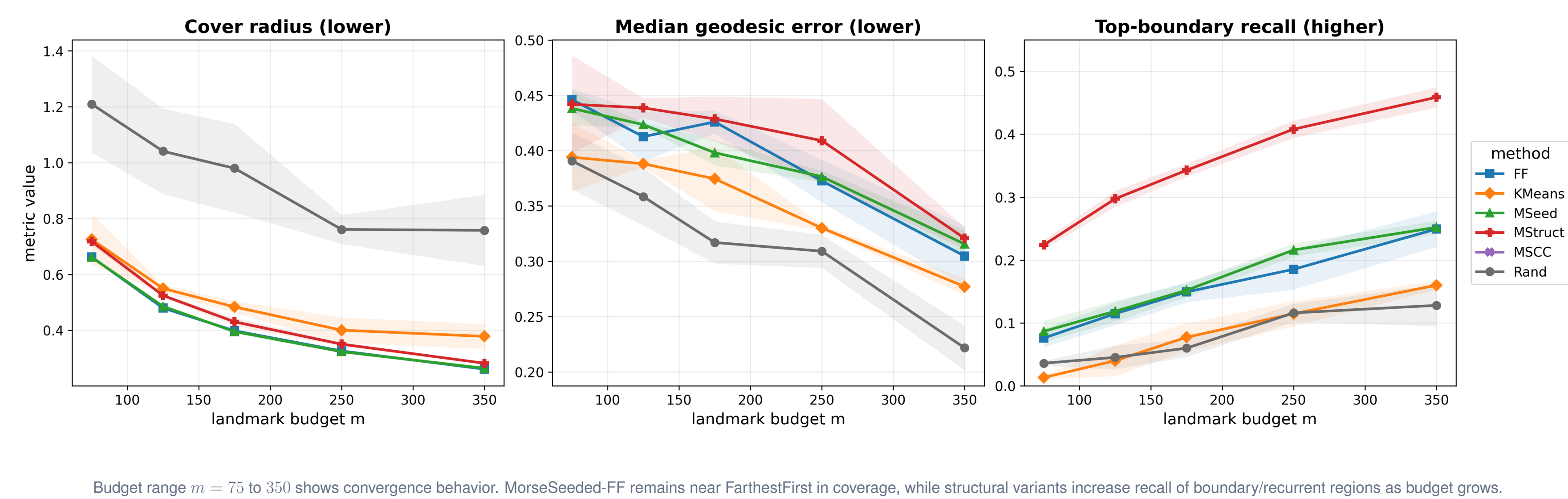
Means across seeds. E_{geo} is absolute relative error in kNN shortest-path distance. R_{10} is recall over the top 10% boundary-score nodes. Lower is better for cover/error; higher is better for recall.

5. Quantitative Trade-off: Coverage vs. Structure



MorseSeeded-FF matches FarthestFirst coverage on structured datasets. MorseSCC and MorseStructural increase boundary recall, but pure geodesic distortion is mixed; this is a structural-representation trade-off, not a universal error win.

6. Compression-Budget Sweep: Swiss Roll

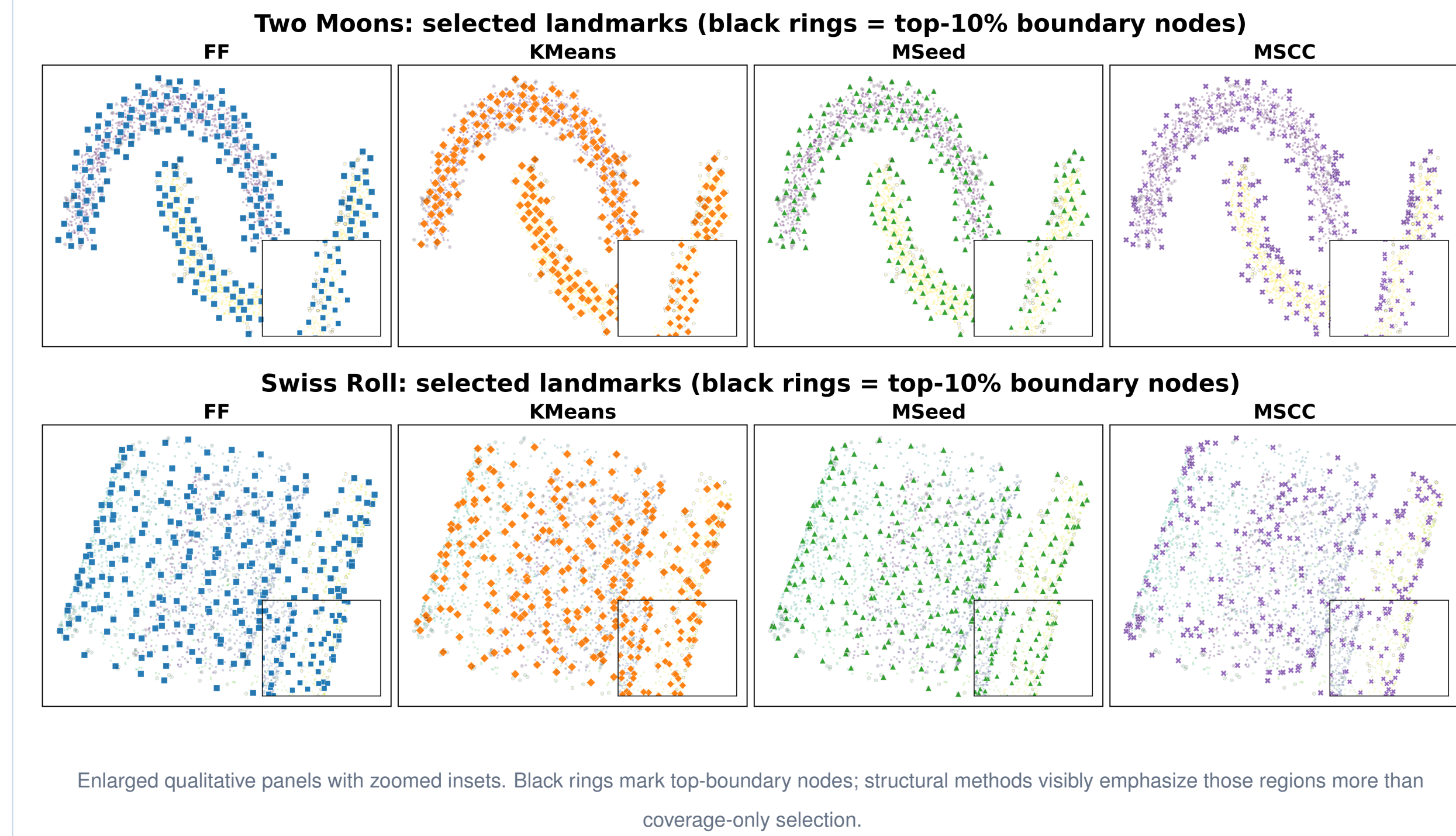


Budget range $m = 75$ to 350 shows convergence behavior. MorseSeeded-FF remains near FarthestFirst in coverage, while structural variants increase recall of boundary/recurrent regions as budget grows.

7. Takeaway in One Sentence

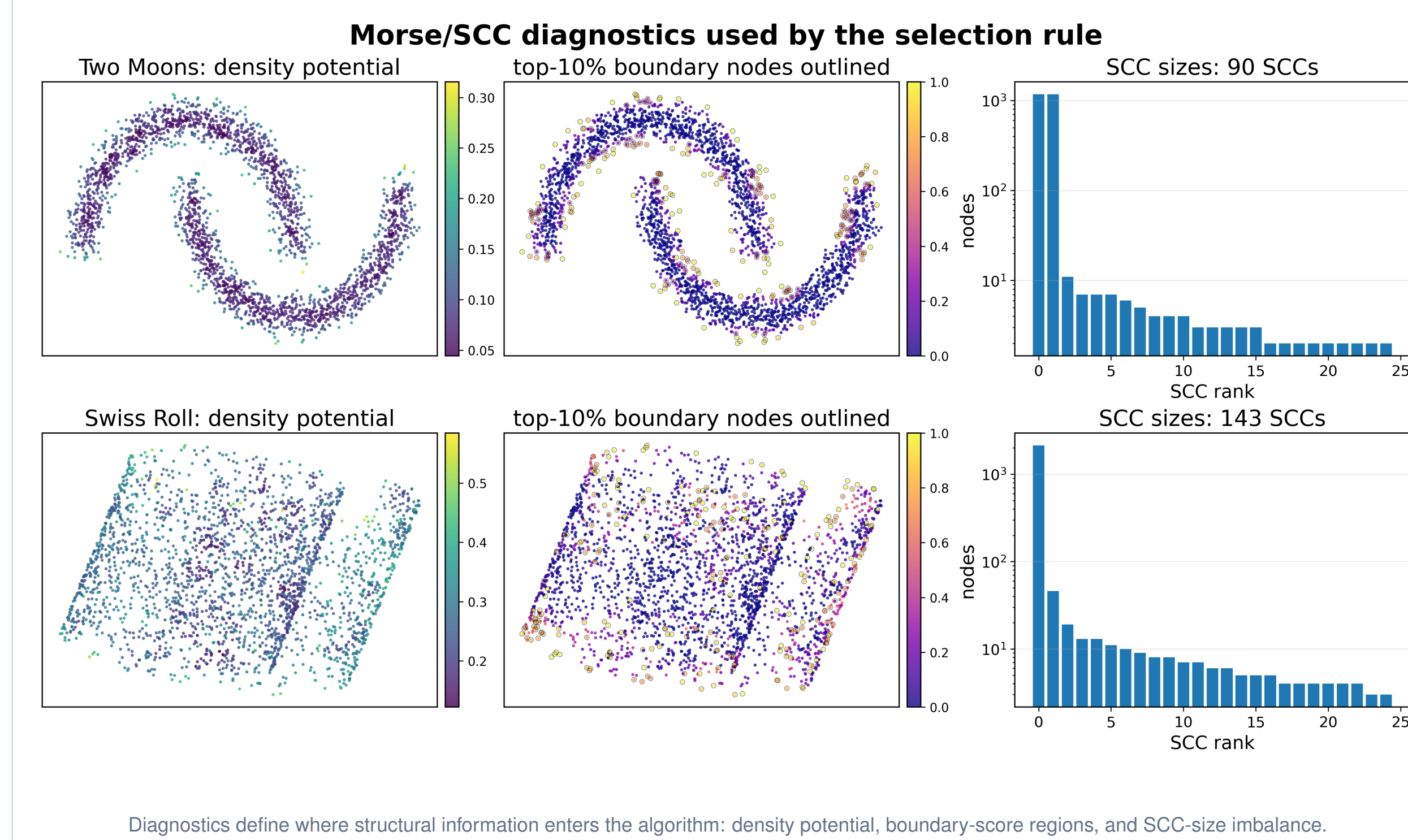
Morse/SCC information is useful when the goal is not merely to minimize Euclidean coverage or average distance error, but to select landmarks that better represent graph-structural roles such as recurrent components and boundary regions.

8. Visual Evidence: Selected Landmarks



Enlarged qualitative panels with zoomed insets. Black rings mark top-boundary nodes; structural methods visibly emphasize those regions more than coverage-only selection.

9. Morse/SCC Diagnostics



Diagnostics define where structural information enters the algorithm: density potential, boundary-score regions, and SCC-size imbalance.

10. Conclusion and Relevance

Primary contribution. We propose **MorseSeeded-FF**, a structure-aware landmarking procedure for kNN graphs. It computes SCCs of a directed density-flow graph, selects boundary-aware seeds from large recurrent regions, and then uses farthest-first traversal to preserve global coverage.

Current evidence. Across Swiss Roll, Two Moons, and Digits PCA, the method preserves FarthestFirst-level coverage on structured manifolds while structural variants improve recall of high-boundary nodes. The distortion results are deliberately reported as mixed: better structural representation does not automatically minimize all geodesic-error metrics. This trade-off clarifies the role of the method: it improves structural representation while retaining competitive coverage, rather than claiming a universal win on every distortion metric.

11. Final Research Narrative

Contribution

We formulate structure-preserving landmarking as a joint problem: preserve k-center coverage while sampling nodes that play important roles in the graph induced by the data manifold.

Empirical signal

MorseSeeded-FF matches FarthestFirst coverage on Swiss Roll and Two Moons, while SCC-aware variants substantially increase recall of top-boundary nodes.

Limitation and next step

The current method exposes a coverage-structure trade-off. Next, we will replace the coarse SCC proxy with multi-scale Morse summaries and adaptive budget allocation.

